

Software Test Plan (STP)

Project: Hospital Management System

Version: 1.0

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1. Introduction

Purpose

This document defines the test plan for the Hospital Management System (HMS) v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing the CLI-based, C / C++ / MySQL application defined in the HMS SRS v1.0 and SAD v1.0.

Scope

Testing covers HMS features accessible through the command-line interface: authentication & access control, patient registration & records, appointment scheduling, billing & invoicing, doctor/staff management, electronic prescriptions, visit summaries, and operational reporting. MySQL server internals, OS-level networking/TLS libraries, and workstation hardware drivers are excluded, as is any integration with external insurance clearing houses or national health information exchanges (out of scope per SRS §1.2).

References

HMS Software Requirements Specification (SRS) v1.0; HMS Software Architecture and Design Specification (SAD) v1.0; applicable healthcare data-protection regulations (HIPAA-equivalent).

Definitions

HMS (Hospital Management System), SRS (Software Requirements Specification), SAD (Software Architecture and Design Specification), RTM (Requirements Traceability Matrix), RBAC (Role-Based Access Control), PHI/PII (Protected Health Information / Personally Identifiable Information), CLI (Command-Line Interface), UAT (User Acceptance Testing).

2. Test Items

- Authentication & Access Control phase (login, RBAC, account logout)
- Patient Registration & Records phase
- Appointment Scheduling phase
- Billing & Invoicing phase
- Doctor / Staff Management & Prescription phase
- Reporting phase
- Data Access Layer (MySQL queries, transaction locking, audit logging)
- CLI Interface Layer (menu navigation, input validation, error messages)

3. Features to be Tested

Features mapped to SRS requirement IDs:

Functional (HMS-F-###)

- HMS-F-001: Authenticate user via username/password
- HMS-F-002: Enforce role-based access control (Admin, Doctor, Receptionist, Billing Staff)
- HMS-F-003: Lock account after 5 consecutive failed login attempts
- HMS-F-010: Register new patient with unique Patient ID
- HMS-F-011: Search patient records by ID, name, or phone
- HMS-F-012: View/update patient medical history, diagnoses, prescriptions
- HMS-F-013: Maintain full audit trail of record changes
- HMS-F-020: Book appointment with a specific doctor/slot
- HMS-F-021: Prevent double-booking of a doctor time slot
- HMS-F-022: Cancel or reschedule an appointment
- HMS-F-030: Generate itemized invoice for a visit
- HMS-F-031: Record payments against an invoice
- HMS-F-040: Manage doctor/staff profiles and schedules
- HMS-F-042: Issue electronic prescription linked to patient record
- HMS-F-043: Generate and export patient visit summary

Non-Functional (HMS-NF-###)

- HMS-NF-001: Appointment search/booking response time $\leq 2s$ for 95% of transactions
- HMS-NF-002: High availability during hospital operating hours
- HMS-NF-003: PHI/PII protected; no sensitive data stored in plaintext
- HMS-NF-004: Audit logs retained for a minimum of 7 years
- HMS-NF-005: UI operable by staff within 30 minutes of training (usability test)

Security (HMS-SR-###)

- HMS-SR-001: TLS 1.2+ mandatory for client–MySQL server connections
- HMS-SR-002: Passwords stored using salted cryptographic hash
- HMS-SR-003: RBAC enforced at both CLI and data-access layer
- HMS-SR-004: Session timeout after 15 minutes of inactivity
- HMS-SR-005: Sensitive patient identifiers stored as one-way hash

4. Features Not to be Tested

- External insurance clearing-house / national health information exchange integrations (out of scope per SRS §1.2)
- MySQL server internals / storage-engine correctness (assumed reliable; DBMS vendor responsibility)

- OS-level networking and TLS library internals (assumed correct; only HMS's usage of TLS is verified)
- Workstation hardware/driver-level behavior beyond basic keyboard/console I/O
- Third-party notification gateways (out of scope per SRS)

5. Test Approach / Strategy

Levels

- Unit tests: individual C / C++ functions within each phase (e.g., `register_patient()`, `book_appointment()`, password-hashing routines)
- Integration tests: phase-to-phase function calls and the Data Access Layer's interaction with MySQL
- System tests: end-to-end CLI workflows (e.g., login → register patient → book appointment → generate invoice)
- Acceptance tests (UAT): validated against SRS acceptance criteria by course evaluators / role-representative reviewers

Types

- Functional testing (core feature behavior against each HMS-F requirement)
- Regression testing (re-run affected suites after each module change)
- Performance testing (response time per HMS-NF-001; concurrent-booking load per HMS-F-021)
- Usability testing (30-minute training / task-completion test per HMS-NF-005)
- Security testing (STRIDE-based checks per SAD §3.9; see §5.1 below)

Entry Criteria

A stable CLI build compiles without errors; the MySQL test schema is deployed and seeded with test data; the test environment (compiler toolchain, MySQL client) is ready.

Exit Criteria

100% of planned test cases executed; 0 critical or high-severity defects open; all high-priority functional and security acceptance criteria satisfied; the RTM (§13) shows full requirement coverage.

5.1 Security Validation

- Verify passwords and sensitive patient identifiers are stored as salted one-way hashes, never in plaintext (HMS-SR-002, HMS-SR-005) — via direct database inspection.
- Verify TLS 1.2+ is enforced on the CLI-to-MySQL connection; connections below TLS 1.2 are refused (HMS-SR-001).
- Verify RBAC is enforced at both the CLI menu level and the data-access layer by attempting out-of-role actions (HMS-SR-003).
- Verify automatic session timeout after 15 minutes of inactivity, requiring re-authentication (HMS-SR-004).

- Verify account lockout after 5 consecutive failed login attempts, with an audit event logged (HMS-F-003).
- Verify audit-log completeness and tamper-resistance for all record create/update/delete actions (HMS-F-013, HMS-NF-004).
- Fuzz-test CLI input fields (patient data, login fields) for unexpected input, buffer-handling issues, and crashes, given C / C++ manual memory management.

6. Test Environment

Hardware

Standard desktop/laptop workstation(s) capable of running the compiled CLI executable, on Windows or Linux (per SRS §2.4).

Software

HMS CLI application build; MySQL Server (dedicated test instance) with seeded schema; C / C++ compiler and build toolchain (e.g., GCC/Clang, CMake).

Tools

- Unit testing: Google Test / Unity (C / C++)
- CLI automation: Python (subprocess) or expect scripts to drive CLI input/output and assert results
- Database verification: MySQL client / MySQL Workbench
- Performance / load: a custom concurrency test harness simulating simultaneous login/booking requests (JMeter/Postman are not applicable — HMS exposes no HTTP API)
- Memory / defect analysis: Valgrind or AddressSanitizer
- Defect tracking: Jira or GitHub Issues

Test Data

Dummy patient demographics; sample doctor schedules and appointment slots; sample invoices/payments; seeded login accounts for each role (Receptionist, Doctor, Administrator, Billing Staff).

7. Test Schedule

Milestones, finalized against the team's project timeline (test case design complete as of 16-Sep-2026; final submission targeted ahead of the 2nd week of November 2026):

Milestone	Target Date
Test case design	16-Sep-2026
Environment setup (MySQL test instance, toolchain)	14-Oct-2026
Unit test execution	21-Oct-2026
Integration & system test execution	30-Oct-2026

Milestone	Target Date
UAT	03-Nov-2026
Test Summary Report / Final submission	05-Nov-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Aniket Raosaheb Jadhav	Prepare plan, coordinate execution
Test Engineer	Aniket S Patil	Design & execute test cases, log defects
Developer	Anirudh Nagarajan, Anirudha Alagawadi	Support defect fixes and triage
Course Coordinator	Asha Kumari	Approve test results, sign off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable CLI build delivery	Request early smoke-test builds from the dev team
MySQL test environment downtime / misconfiguration	Maintain a backup local MySQL instance (e.g., a Docker container) for testing
Manual memory-management bugs (C / C++) causing crashes mid-test	Run the test suite regularly under Valgrind / AddressSanitizer
Limited availability of test workstations	Prioritize core-flow testing first; schedule shared machine usage

11. Assumptions & Dependencies

- The MySQL test instance will be available and stable throughout test execution.
- Seeded test data (patients, doctors, schedules) will be provided before execution begins.
- The development environment (compiler, libraries) will be consistent across all test machines.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable for more than 4 hours.
- The build is too unstable, blocking more than 30% of planned test cases.
- A critical crash or data-corruption defect is found.

Resume testing if:

- Blocking defects are resolved and verified.
- The environment is stabilized.

13. Test Case Management & Traceability

The RTM in SRS §8 maps every requirement to its test case ID(s); the excerpt below shows the convention used across all modules.

Req ID	Requirement (short)	Test case(s)
HMS-F-001	Authenticate user	TC-Auth-01
HMS-F-002	Role-based access	TC-Auth-02
HMS-F-010	Register patient	TC-Reg-01
HMS-F-012	Update medical record	TC-Rec-01
HMS-F-020	Book appointment	TC-Appt-01
HMS-F-021	Prevent double-booking	TC-Appt-02
HMS-F-030	Generate invoice	TC-Bill-01
HMS-F-042	Issue prescription	TC-Rx-01
HMS-F-043	Generate visit summary	TC-Doc-01
HMS-NF-001	Response time target	TC-Perf-01
HMS-NF-005	Usability / training time	TC-Usab-01
HMS-SR-002	Password hashing	TC-Sec-03
HMS-SR-004	Session timeout	TC-Sec-05
HMS-SR-005	Sensitive ID hashing	TC-Sec-06

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed / failed
- Defect density
- Defect aging
- Requirement coverage (against the RTM)

Reports:

- Daily execution status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Aniket Raosaheb Jadhav	16-Sep-2026
Dev Lead	Anirudh Nagarajan	16-Sep-2026
Course Coordinator	Asha Kumari	16-Sep-2026